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MID SEMESTER EXAMINATION MARCH - 2025

Name of the Program: **B.Tech., ME**Semester: **8th**Name of the Course: **VEHICLE BODY SAFETY AND CONTROL**Course Code: **(TAE 801)**

Time: 90 Minutes

Maximum Marks: 50

Note:-

- (i) Answer **all the questions** by choosing **anyone of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10marks)	CO1
(a)	Define active and passive safety in the context of automotive safety systems.	
OR		
(b)	List the different types of impact tests performed on vehicles to assess crashworthiness.	
Q2	(10 marks)	CO2
(a)	Explain the significance of crumple zones in vehicle crash safety and how they contribute to energy absorption.	
OR		
(b)	Describe the role of airbags in occupant protection and how they function in a crash event.	
Q3	(10 marks)	CO3
(a)	Illustrate how electric power-assisted steering improves active safety in modern vehicles.	
OR		
(b)	Demonstrate how seat belts contribute to reducing occupant injuries during a rollover crash test.	
Q4	(10 marks)	CO4
(a)	Compare and contrast the impact behavior of a small car versus a larger vehicle in crash testing.	
OR		
(b)	Analyze the relationship between vehicle structure stiffness and energy absorption characteristics in crash safety.	
Q5	(10 marks)	CO5
(a)	Evaluate the effectiveness of pedestrian protection measures in modern vehicle design.	
OR		
(b)	Propose a new or improved structural safety feature that could enhance occupant safety in small cars.	

